

ECON 108

INTRODUCTION TO ECONOMICS

INSTRUCTOR: ALEJANDRO MELO PONCE

Measuring the economy's performance

CHAPTER VI

CHAPTER OUTLINE

The Simple Circular Flow

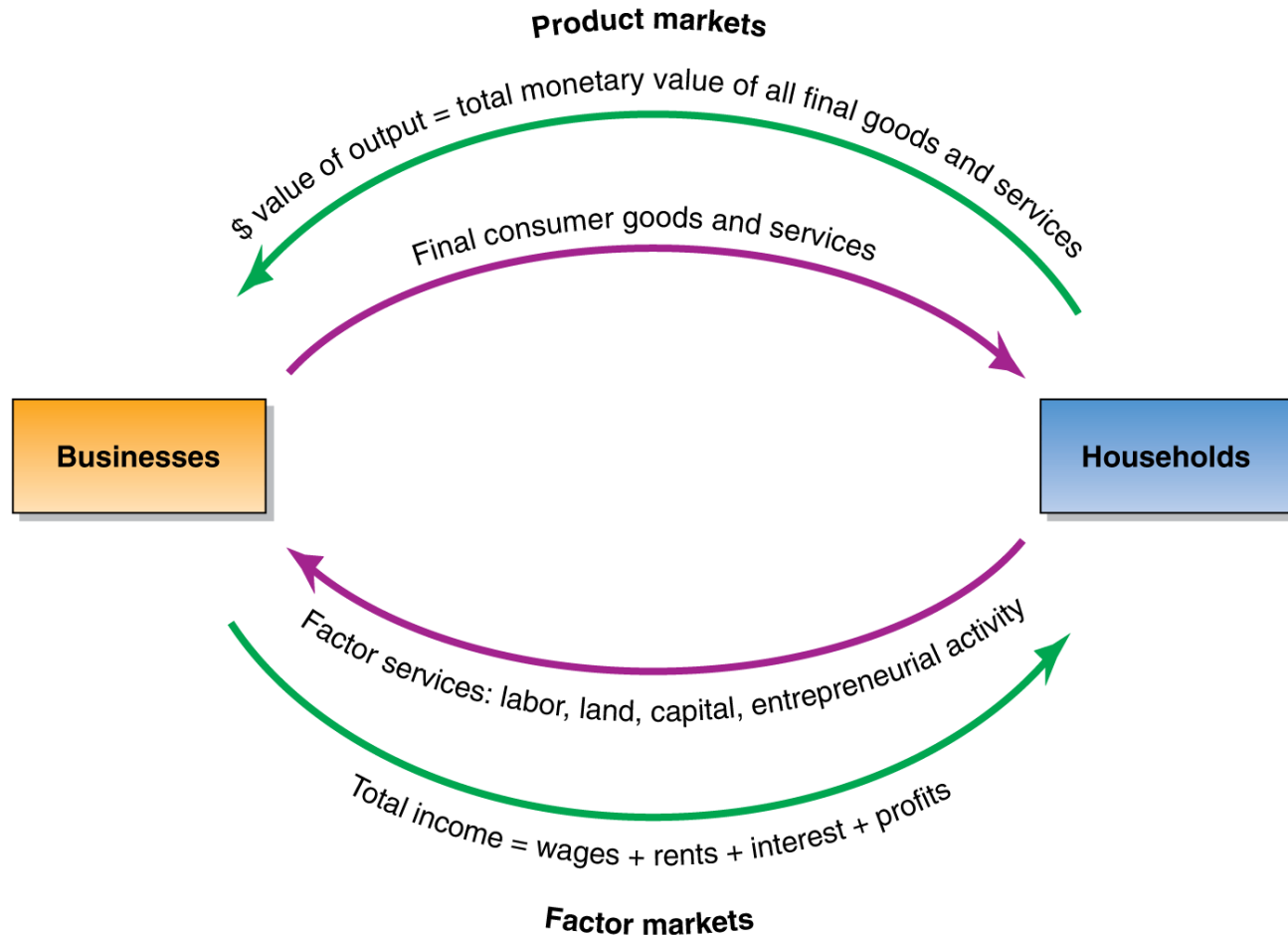
National Income Accounting

Two Main Methods of Measuring GDP

Real Variables versus Nominal Variables

Comparing GDP Throughout the World

THE SIMPLE CIRCULAR FLOW



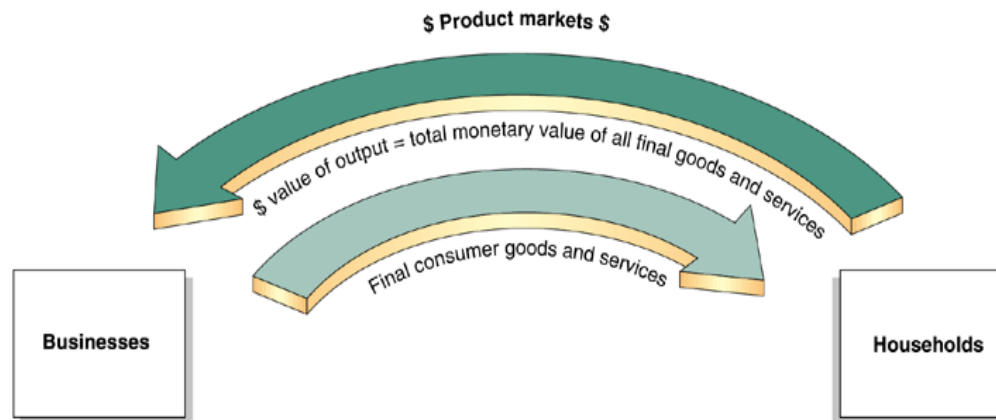
THE SIMPLE CIRCULAR FLOW

- In every economic exchange, the seller receives exactly the same amount that the buyer spends.
- Goods and services flow in one direction and money payments flow in the other.

THE SIMPLE CIRCULAR FLOW

Product Markets

Transactions in which households buy goods

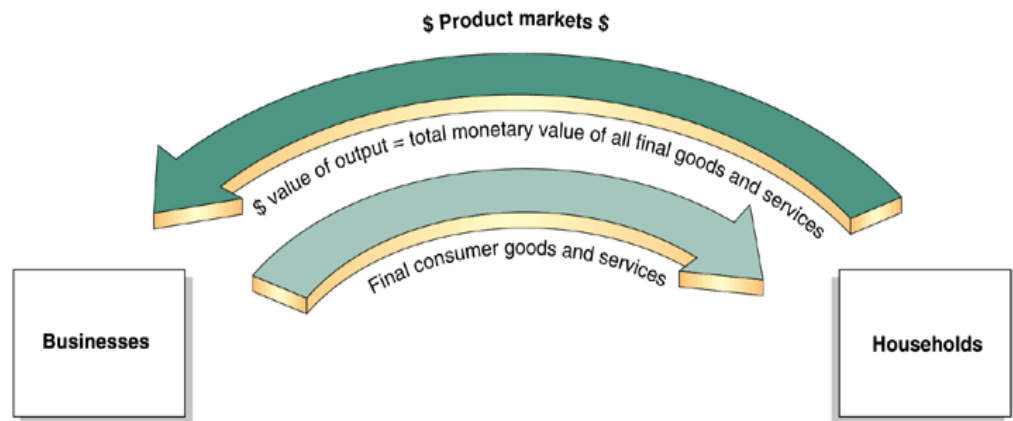


\$ Factor markets \$

THE SIMPLE CIRCULAR FLOW

Final Goods and Services

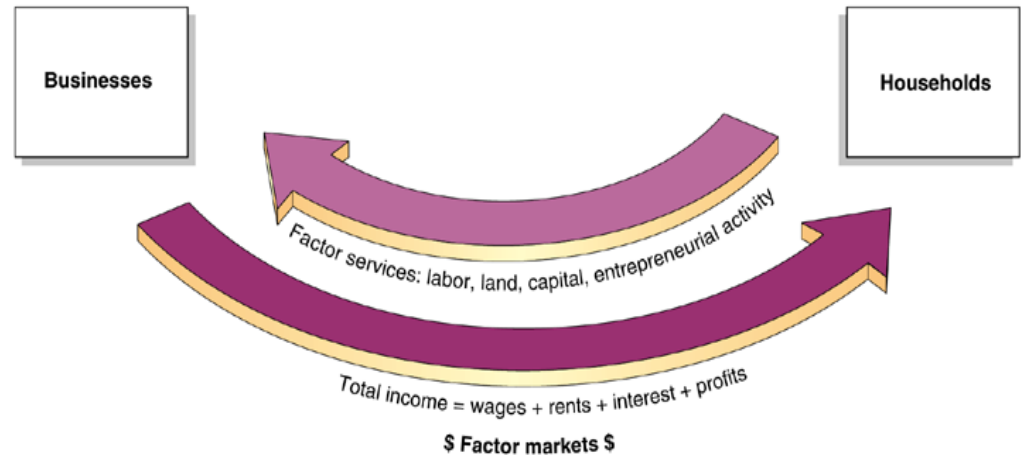
Goods and services that are at their final stage of Production and will not be transformed into yet other goods or services



THE SIMPLE CIRCULAR FLOW

Factor Markets

Transactions in which businesses buy resources

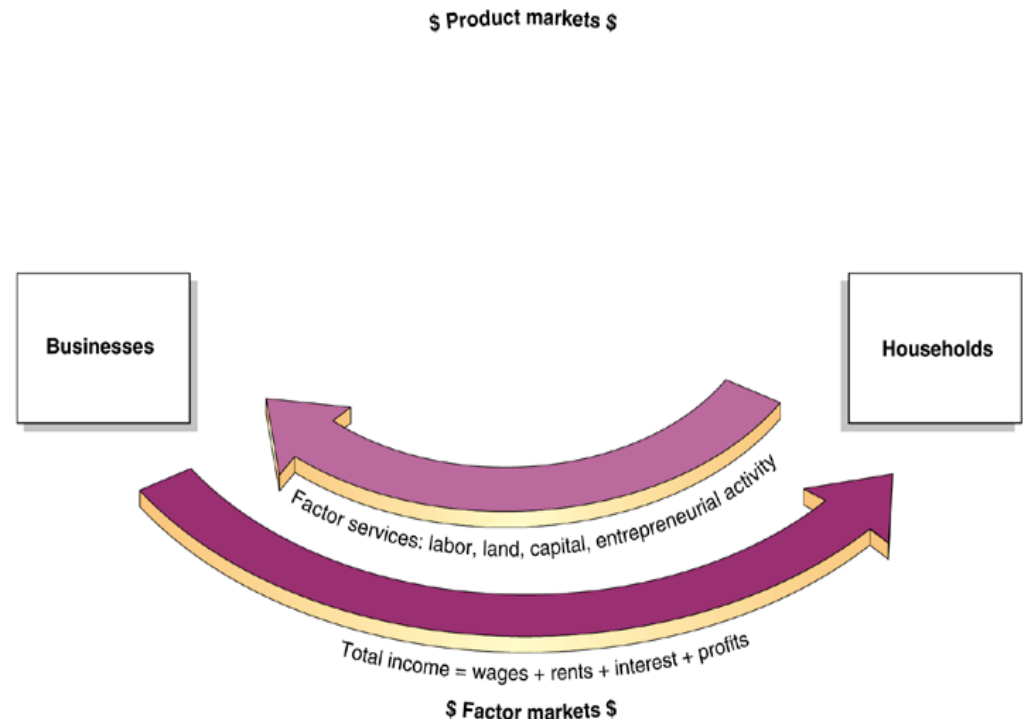


THE SIMPLE CIRCULAR FLOW

Total Income

The yearly amount earned by the nation's factors of production.

The annual cost of producing the entire output of **final goods and services**.



THE SIMPLE CIRCULAR FLOW

Question:

- Why must total income be identical to the dollar value of total output?

Answer:

- Every transaction simultaneously involves an expenditure and a receipt.
- Profit is defined as what is *left over* from total business receipts after all other costs have been paid.

Be ready for the test!

The observation that goods and services flow in one direction and money payments flow in another direction is the principle behind

- a. the double coincidence of wants.
- b. a barter economy.
- c. a pure command economy.
- d. the circular flow of income.

Be ready for the test!

According to the circular flow, the dollar value of a nation's output is equal to

- a. profits.
- b. total income.
- c. quantity of money.
- d. wages.

Be ready for the test!

Which of the following statements is FALSE?

- a. One definition of total income is that it is the annual cost of producing the entire output of final goods and services.
- b. Transactions in which households buy final goods and services occur in the factor market.
- c. The value of total output is identical to total income.

Be ready for the test!

In the circular flow model, profits are considered to be

- a. a subtraction from the GDP.
- b. a cost of doing business.
- c. equal to zero, or else the circular flow would be out of balance.
- d. a form of interest payment.

NATIONAL ACCOUNTING

Gross Domestic Product (GDP)

- The total **market value** of all **final** goods and services produced by factors of production located **within** a nation's borders

NATIONAL ACCOUNTING

Observations

- The GDP measures
 - the dollar value of final output
 - produced during a year
 - by factors of production located within the country

NATIONAL ACCOUNTING

Intermediate Goods

- Goods used up entirely in the production of final goods.
- GDP does not count intermediate goods because to do so would be to count twice.
- Example: grain and bread.

NATIONAL ACCOUNTING

Financial Transactions

- Are not included in GDP calculation
- Represent ownership transfers not value added to the economy

NATIONAL ACCOUNTING

Financial Transactions and Transfer Payments

- Securities
 - Stocks and bonds
- Government transfer payments
 - Social Security
 - Unemployment compensation
- Private transfer payments
 - Individual or corporate gifts

NATIONAL ACCOUNTING

Second hand goods

- Second hand goods are not included in the calculation of the GDP. *Why?*

NATIONAL ACCOUNTING

Other excluded transactions

- Household production
- Legal underground transactions
- Illegal underground transactions

NATIONAL ACCOUNTING

GDP's limitations

- Excludes non-market production
- Quality of life is not measured
- GDP poorly measures a nation's well-being
- GDP is not always comparable among different countries

NATIONAL ACCOUNTING

GDP's limitations

- GDP **is** a measure of the value of production in terms of market prices and an indicator of economic activity.
- GDP **is not** a measure of nation's overall welfare.

NATIONAL ACCOUNTING

Value Added

- The amount of dollar value contributed to a product at each stage of its production.
- Value added is the dollar value of an industry's sales minus the value of intermediate goods used in production.

NATIONAL ACCOUNTING

Value Added: Example

Stage of Production	Dollar Value of Sales		Value Added
Stage 1: Fertilizer and Seed	\$.03		\$.03
Stage 2: Growing	.06		\$.03
Stage 3: Milling	.12		\$.06
Stage 4: Baking	.30		\$.18
Stage 5: Retailing	.45		\$.15
Total dollar value of all sales	\$.96	Total value added	\$.45

TWO MAIN METHODS OF COMPUTING GDP

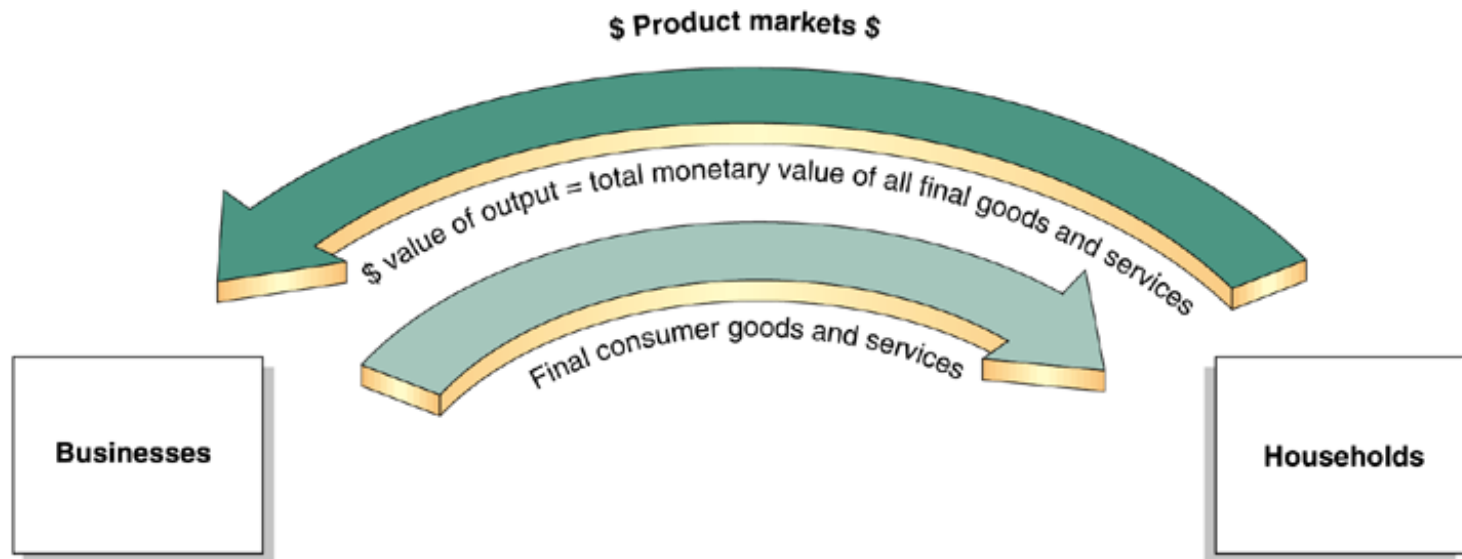
Expenditure Approach

- Computing GDP by adding up the dollar value at current market prices of all final goods and services

Income Approach

- Measuring GDP by adding up all components of national income, including wages, interest, rent, and profits

TWO MAIN METHODS OF COMPUTING GDP



Expenditure Approach

TWO MAIN METHODS OF COMPUTING GDP

Expenditure Approach

- Mathematical representation using the expenditure approach

$$GDP = C + I + G + NX$$

Expenditure Approach

Consumption Expenditure

Durables

- Life span of more than three years

Non-durables

- Life span of less than three years

Services

- Intangible commodities

Expenditure Approach

Gross Private Domestic Investment

The creation of capital goods, such as factories and machines, that can yield production and hence consumption in the future

- Fixed investment
- Inventory investment
- New residential structure

Expenditure Approach

Gross Private Domestic Investment

- **Fixed investment:** purchases by businesses of newly produced capital goods, such as production machinery and office equipment.
- **Inventory investment:** changes in the stocks of finished goods and goods in process, as well as changes in the raw materials that businesses keep on hand.
- **New residential structures:** new housing.

Expenditure Approach

Gross Private Domestic Expenditure

- The layperson's notion of investment often relates to the purchase of stocks and bonds.
- For our purposes, such transactions simply represent the *transfer of ownership* of assets called stocks and bonds.
- Thus, you must keep in mind the fact that in economics, **investment** refers only to **additions to productive capacity**, not to transfers of assets.

Expenditure Approach

Government Expenditure

- State, local, and Federal
- Valued at cost

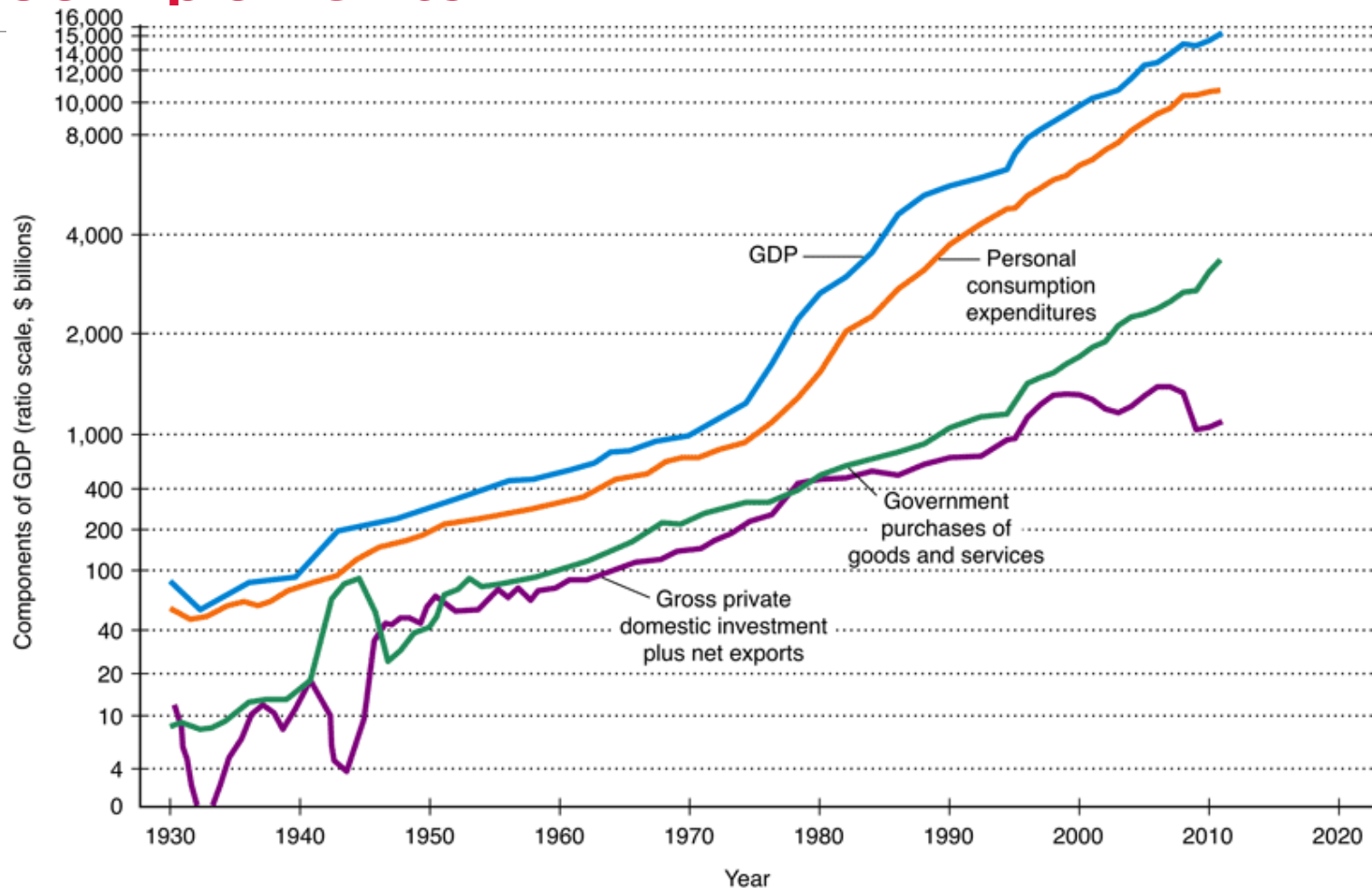
Expenditure Approach

Net Exports

- Net expenditures from the foreign sector

Net exports (NX) = total exports - total imports

Expenditure Approach: GDP components



Gross Variables versus Net Variables

Depreciation

- Reduction in the value of capital goods over a one-year period due to physical wear and tear, and also to obsolescence

$$\text{NDP} = \text{GDP} - \text{depreciation}$$

Gross Variables versus Net Variables

Some formulas

$$GDP = C + I + G + NX$$

$$NDP = C + I + G + NX - \text{depreciation}$$

$$\text{Net Investment} = I - \text{depreciation}$$

$$NDP = C + \textit{net } I + G + NX$$

Be ready for the test!

If GDP equals \$1 trillion, gross private investment expenditures are \$200 billion, exports equal imports, and government spending is \$400 billion, then

- a. consumption expenditures are \$200 billion.
- b. consumption expenditures are \$400 billion.
- c. spending on consumer durables must be \$400 million.
- d. we cannot determine what expenditures on consumption are without more information.

Be ready for the test!

An example of an increase in gross private domestic investment spending that also increases GDP is when

- a. a family sells its home because of a transfer.
- b. a farmer buys a used tractor.
- c. inventories of new cars accumulate on the lots of car dealers.
- d. government increases spending on infrastructure.

Be ready for the test!

When the total quantity of unsold new Chevrolets increases from 50,000 to 75,000

- a. the economy experiences a period of disinvestment.
- b. the economy's level of competition increases.
- c. there is an increase in the rate of aggregate output.
- d. the producer engages in an inventory investment.

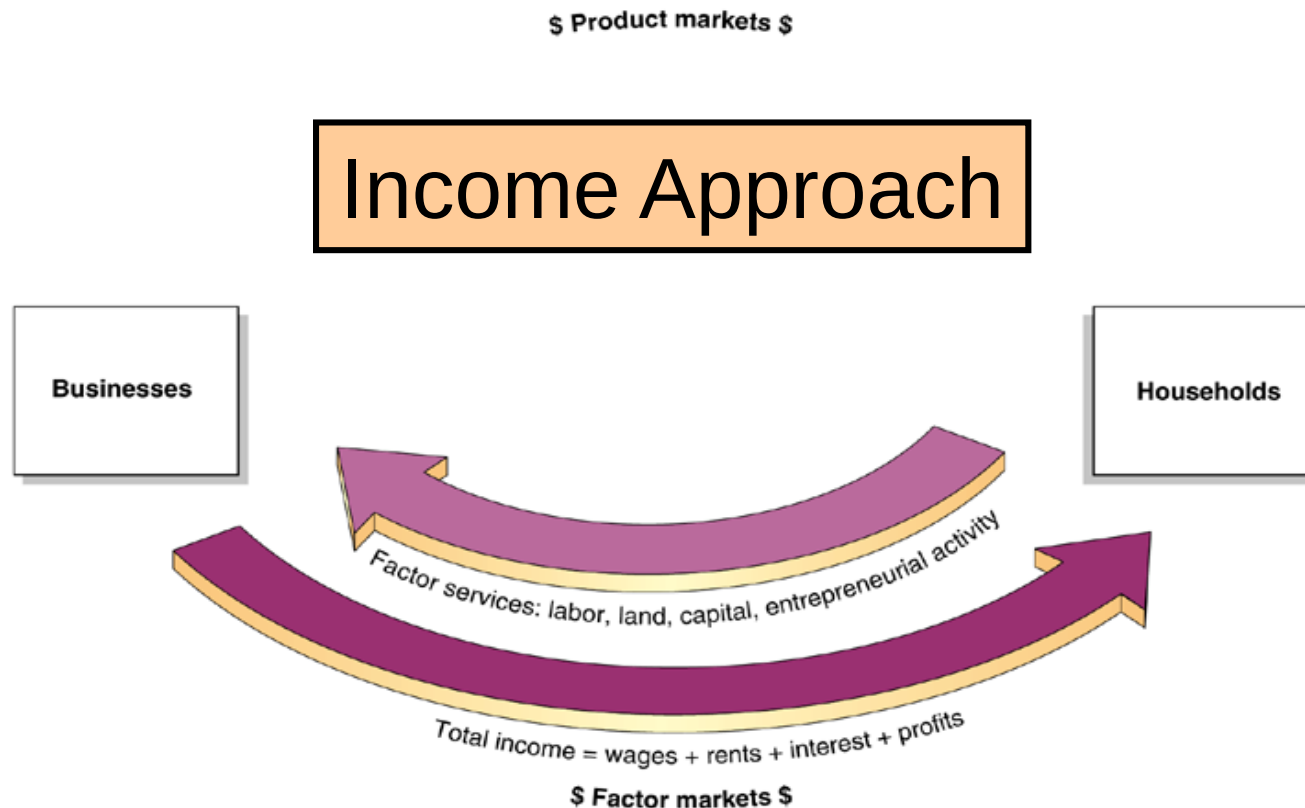
Income Approach

Gross Domestic Income

- The sum of all income—wages, interest, rent, and profits—paid to the four factors of production
- We also need to add the value of indirect taxes paid by households and depreciation.

Income Approach

Gross Domestic Income



Income Approach

Gross Domestic Income

Income expense items

- Wages
- Interest
- Rent
- Profits

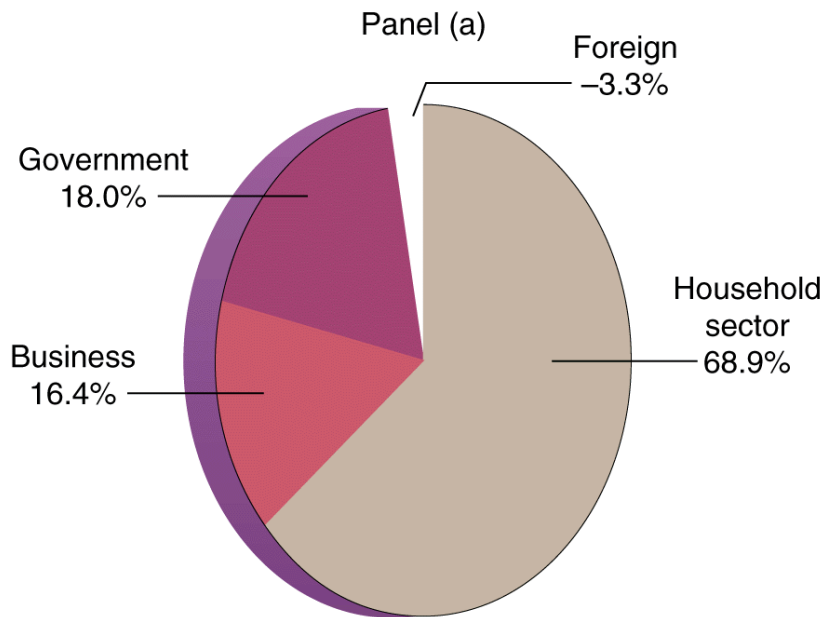
Income Approach

Gross Domestic Income

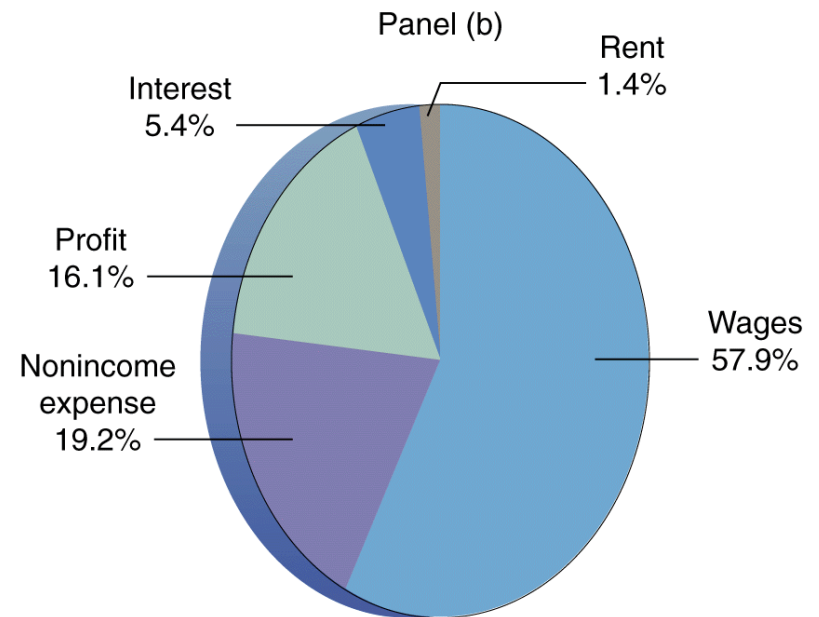
Non Income expense items

- **Indirect Business Taxes:** All business taxes except the tax on corporate profits (i.e., sales and business property taxes)
- **Depreciation**

Gross Domestic Product and Gross Domestic Income, 2003 (in billions of 2003 dollars per year)



Expenditure Approach



Income Approach

Other Components of National Income Accounting

National Income (NI)

- The total of all factor payments to resource owners

Personal Income (PI)

- The amount of income that households actually receive before they pay personal income taxes

Disposable Personal Income (DPI)

- Personal income after personal income taxes have been paid

Going from GDP to Disposable Income, 2011

	Billions of Dollars
Gross domestic product (GDP)	15,067.5
Minus depreciation	<u>-2,210.2</u>
Net domestic product (NDP)	12,857.3
Minus indirect business taxes and transfers	-1,374.4
Plus other business income adjustments	1,296.7
Plus net U.S. income earned abroad	<u>71.0</u>
National income (NI)	12,850.6
Minus corporate taxes, Social Security contributions, corporate retained earnings	-1,881.5
Plus government transfer payments	<u>1,787.2</u>
Personal income (PI)	12,756.3
Minus personal income taxes	<u>-1,132.3</u>
Disposable personal income (DPI)	11,624.0

Sources: U.S. Department of Commerce and author's estimates.

Be ready for the test!

When using the income approach to estimate GDP, which of the following would NOT be included?

- a. The implicit rental value of owner-occupied housing
- b. Proprietors' income
- c. Personal consumption expenditures
- d. Net interest received by households

Be ready for the test!

Which of the following is included in the calculation of GDP using the income approach?

- I. Wages and salaries.
- II. Net exports.

- a. I only
- b. II only
- c. Both I and II
- d. Neither I nor II

REAL VARIABLES VERSUS NOMINAL VARIABLES

Nominal Values

- Measurements in terms of the actual market prices at which goods are sold; expressed in *current* dollars.

Real Values

- Measurements after adjustments have been made for changes in the average of prices between years; expressed in *constant* dollars

REAL VARIABLES VERSUS NOMINAL VARIABLES

Example

Base Year = 1992

Price Index = 100

REAL VARIABLES VERSUS NOMINAL VARIABLES

Example

$$\text{Real 1992 GDP} = \frac{\text{nominal GDP}}{\text{price index}} \times 100$$

$$\text{Real 1992 GDP} = \frac{\$6,020.2}{100} \times 100 = \textbf{\$6,020.2 billion}$$

Real GDP = nominal GDP in the
base year

REAL VARIABLES VERSUS NOMINAL VARIABLES

Example

- Year 1993
- Price Index = 102.2

$$\text{Real 1993 GDP} = \frac{\$6,343.3}{102.2} \times 100 = \$6,206.8 \text{ billion}$$

REAL VARIABLES VERSUS NOMINAL VARIABLES

Example

- Year 1987
- Price Index = 82.7

$$\text{Real 1987 GDP} = \frac{\$4,539.9}{82.7} \times 100 = \text{\$5,489.6 billion}$$

REAL VARIABLES VERSUS NOMINAL VARIABLES

Questions

- Why is the price index greater than 100 for 1993?
- Why is the price index less than 100 for 1987?
- Why is Real GDP greater than Nominal GDP for 1987 and less than Nominal GDP for 1993?

REAL VARIABLES VERSUS NOMINAL VARIABLES

GDP deflator

- Nominal (current) dollars GDP
- Real (constant) dollars GDP

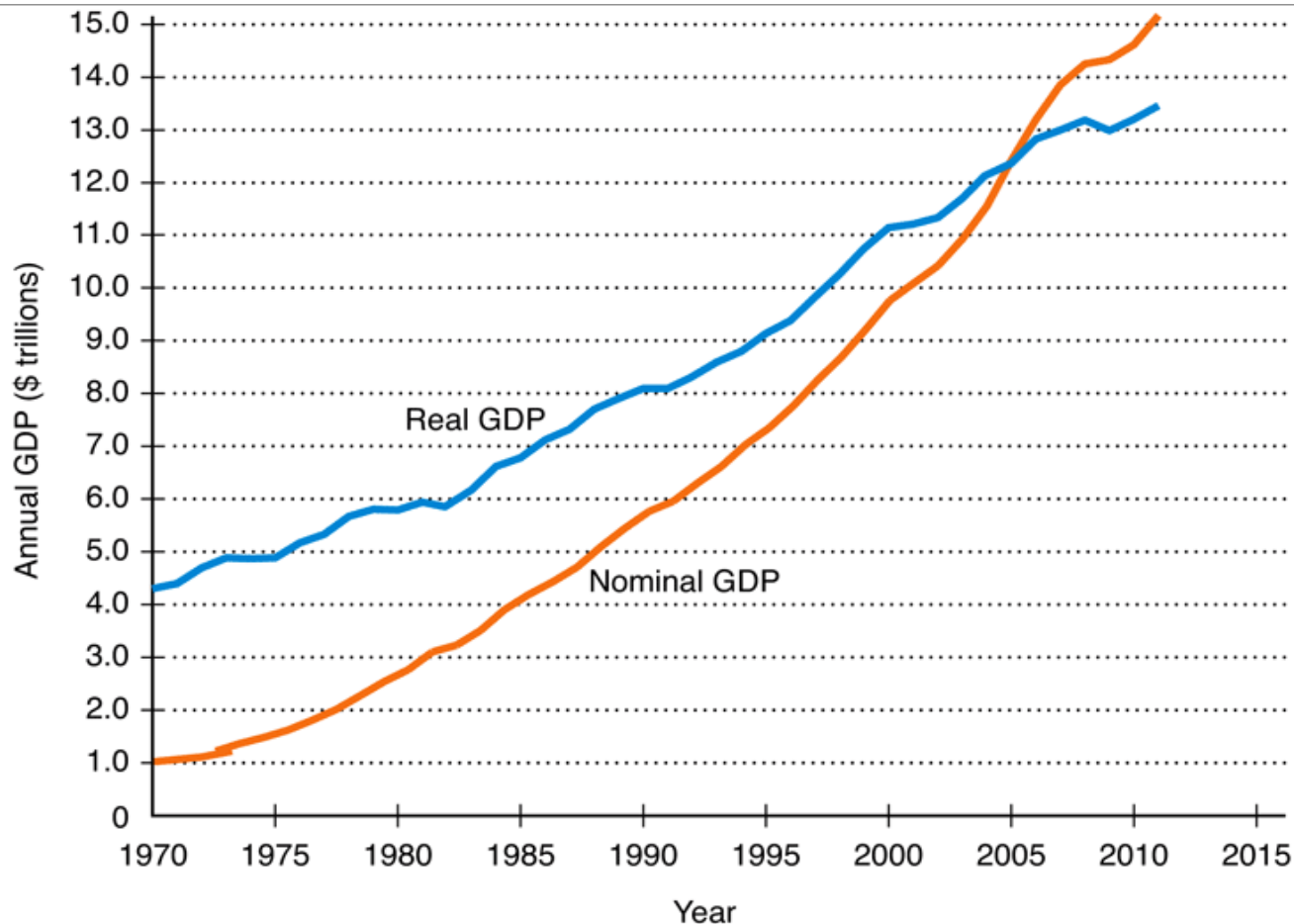
$$\text{GDP deflator} = \frac{\text{nominal GDP}}{\text{real GDP}} \times 100$$

REAL VARIABLES VERSUS NOMINAL VARIABLES

(1) Year	(2) Nominal GDP (billions of dollars per year)	(3) Price Index (base year 2005 = 100)	(4) = [(2) ÷ (3)] × 100 Real GDP (billions of dollars per year, in constant 2005 dollars)
2001	10,286.2	90.654	11,347.2
2002	10,642.3	92.113	11,553.0
2003	11,142.1	94.099	11,840.7
2004	11,867.8	96.769	12,263.8
2005	12,638.4	100.000	12,638.4
2006	13,398.9	103.263	12,976.2
2007	14,077.6	106.221	13,254.1
2008	14,441.4	108.481	13,312.1
2009	14,256.3	109.745	12,987.4
2010	14,635.0	110.952	13,190.4
2011	15,067.5	112.395	13,405.8

Sources: U.S. Department of Commerce, Bureau of Economic Analysis, and author's estimates.

REAL VARIABLES VERSUS NOMINAL VARIABLES



Source: U.S. Department of Commerce.

Be ready for the test!

The nominal value of GDP is expressed in

- a. constant relative prices.
- b. constant dollars.
- c. current dollars.
- d. inflation-adjusted prices.

Be ready for the test!

If nominal GDP in 2005 was \$3,000 billion, and the price level index was 330 (2002 = 100), then real GDP was about

- a. \$105 billion.
- b. \$4,220 billion.
- c. \$909 billion.
- d. \$537 billion.

COMPARING COUNTRIES THROUGHOUT THE WORLD

Adjusting for population differences

Per capita GDP

$$\text{Per capita real GDP} = \frac{\text{real GDP}}{\text{population}}$$

Per capita GDP in U.S. is approximately \$49,000

COMPARING COUNTRIES THROUGHOUT THE WORLD

Adjusting for different currencies

Foreign exchange rate

- Price of one currency in terms of another.
- Example: the exchange rate between U.S. and Argentina is 0.25 because 0.25 dollars are needed in order to buy 1 Argentine peso.
- Per capita GDP in Argentina is 72,808 Argentine pesos (i.e., $0.25 \times 72,808 = \$18,202$)

COMPARING COUNTRIES THROUGHOUT THE WORLD

Purchasing power parity

Adjustment in exchange rate conversions that takes into account differences in the true cost of living across countries

International Example: Purchasing Power Parity Comparisons of World Incomes

The International Monetary Fund accepted the purchasing power parity approach a few years ago

It started presenting the statistics on each country's GDP relative to others and based on the purchasing power parity relative to the dollar.

Comparing GDP Internationally

Country	Annual GDP Based on Purchasing Power Parity (billions of U.S. dollars)	Per Capita GDP Based on Purchasing Power Parity (U.S. dollars)	Per Capita GDP Based on Foreign Exchange Rates (U.S. dollars)
United States	13,886	46,040	46,040
United Kingdom	2,464	34,050	40,660
Germany	3,207	34,740	38,990
France	2,467	33,850	38,810
Japan	4,829	34,750	37,790
Italy	1,988	30,190	33,490
Russia	1,070	14,330	7,530
Brazil	1,122	9,270	5,860
China	3,126	5,420	2,370
Indonesia	373	3,750	1,650

Source: World Bank.

END OF CHAPTER 6...